PREVIEW: built from Claude's suggestions, not author decisions. Not for release.

irAE-Map

Codebook and mapping rationale, version 1.0.0

A clinician-curated crosswalk of immune-related adverse events of immune checkpoint inhibitor therapy to CTCAE v6.0, ICD-10-CM FY2027, and FDA Adverse Event Reporting System reaction terms

[VERIFY: name exactly as it should appear]

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1. Summary

irAE-Map links 36 immune-related adverse event (irAE) categories to the terms and codes that describe them in three vocabularies: the National Cancer Institute's Common Terminology Criteria for Adverse Events (CTCAE) version 6.0, the US diagnosis code set ICD-10-CM (fiscal year 2027), and the MedDRA reaction terms used in the FDA Adverse Event Reporting System (FAERS). Version 1.0.0 contains 906 mappings: 137 to CTCAE terms, 373 to ICD-10-CM code families covering 1,939 billable codes, and 396 to FAERS reaction terms. Each mapping has a role, a confidence rating and a clinical rationale, and each was decided by the author.

CTCAE v6.0, released by NCI in 2026, added terms in part to better describe adverse events of immunotherapy. Of the 134 CTCAE terms in irAE-Map, 29 are new in v6.0. ICD-10-CM gained a dedicated drug-cause code for immune checkpoint inhibitors, T45.AX5-, effective 1 October 2024. Health systems, registries and drug-safety analysts need a public, documented link between these vocabularies; irAE-Map provides one.

2. Sources

Source	Edition	Use in irAE-Map
NCI CTCAE spreadsheet (dctd.cancer.gov)	v6.0, MedDRA 28.0, errata 2026-01-20	Terms, MedDRA LLT codes, grade 1 to 5 definitions, v5 to v6 change flags
CDC/NCHS ICD-10-CM code descriptions and addenda (ftp.cdc.gov)	FY2027, effective 2026-10-01	Codes, billable flags, descriptions, codes new in FY2027
openFDA drug adverse event API (api.fda.gov)	FAERS data updated 2026-07-30; accessed 2026-10-13	Reaction terms reported with the 13 checkpoint inhibitors and report counts

Exact URLs, download dates, file sizes and SHA-256 checksums are in data/raw/PROVENANCE.txt in the repository.

3. Methods

3.1 Category frame

Categories follow the organ groupings of the ASCO 2021 guideline on management of irAEs in patients treated with immune checkpoint inhibitor therapy, with additions where CTCAE v6.0 or the literature identifies a distinct entity. Each category lists the diagnoses most often confused with it (Section 5).

3.2 Candidate generation

- CTCAE v6.0: terms named in each category's rules, matched exactly to the v6.0 spreadsheet; the build fails if a named term is absent.
- ICD-10-CM: codes under named code prefixes, plus codes whose description matches the category's keyword pattern, excluding keyword matches in the neoplasm, pregnancy, perinatal, injury, external-cause and Z chapters. Billable codes are grouped into code families (a named parent code with its children, or the remaining codes in a three-character block) for review, and expanded back to every billable code for release.
- FAERS: for each category keyword, openFDA count queries over reports naming any of the 13 checkpoint inhibitors (290,091 reports) and over all reports (20,692,687), keeping reaction terms that match the category pattern. Reports were matched by openFDA's harmonized generic name or by the raw product name (generic or brand), because the

harmonized field returned no reports for three of the drugs. Each kept term's counts are exact.

- Each candidate received a suggested role, confidence and reason from documented rules (repository file code/categories.py), drafted with AI assistance (Section 8). Rows the rules could not classify were left without a suggestion.

3.3 Author review

The author reviewed every candidate in a workbook and either accepted the drafted role, confidence and reason unchanged, or set them, or excluded the row. Rows not decided were not released. The released table records which (column decision_type), with the reviewer and date.

Role	Meaning
primary	Names the condition itself; enough on its own to identify the category
supporting	A laboratory, symptom, severity or grading term consistent with, but not specific to, the category
lookalike	A different condition often confused with this one; listed so analysts exclude it deliberately
companion	ICD-10-CM drug-cause code that should accompany a condition code

Confidence	Meaning
High	Names the condition and is used for it in practice
Moderate	Right organ and process, but also used for non-immune causes
Low	Supporting or non-specific

3.4 Quality checks

The build verifies that every CTCAE term exists in v6.0 and every released ICD-10-CM code is a valid billable FY2027 code; that every released row has a role, a confidence (except look-alikes) and a rationale; that no suggestion-less row was accepted without a decision; and it reports, for each category, any vocabulary lacking a primary mapping. Results are in data/processed/qa_report.txt.

3.5 FAERS over-representation

For each reaction term, faers_ici_prr = (a / N_ici) / ((b - a) / (N_all - N_ici)), where a is the number of reports naming a checkpoint inhibitor with the term, b the number of all reports with the term, N_ici = 290,091 and N_all = 20,692,687. It describes over-representation only; it is not a disproportionality signal (no deduplication, stratification or confidence interval).

4. Results

Vocabulary	Mappings	Primary	Supporting	Look-alike	Companion
CTCAE v6.0	137	59	64	14	0
ICD-10-CM FY2027	373	143	32	196	2
FAERS MedDRA PT	396	237	102	57	0

Confidence: High 482, Moderate 289, Low 134. 8 terms are primary in more than one category; these are overlap entities such as thyroiditis (both thyroid categories) and the myocarditis-myositis-myasthenia gravis overlap syndrome.

5. Category rationale

For each category: the primary CTCAE terms with their grade 3 definition, primary ICD-10-CM code families, the most-reported primary FAERS terms, and look-alikes. The complete list, including supporting terms, is the released table.

Colitis / enterocolitis / diarrhea (COLITIS)

Organ system: Gastrointestinal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Colitis	Severe abdominal pain; peritoneal signs	High	CTCAE term for colon inflammation; the graded irAE term
Enterocolitis	Severe or persistent abdominal pain; fever; ileus; peritoneal signs	High	Small and large bowel inflammation; ICI enterocolitis

ICD-10-CM	Description	Billable codes	Confidence	Rationale
K52.83*	Microscopic colitis	4	Moderate	Microscopic colitis; reported phenotype of ICI colitis
K52.9	Noninfective gastroenteritis and colitis, unspecified	1	Moderate	Noninfective colitis, unspecified; common but non-specific
K52.89	Other specified noninfective gastroenteritis and colitis	1	Moderate	Other specified noninfective colitis; used for immune-mediated colitis
K52.1	Toxic gastroenteritis and colitis	1	High	Toxic gastroenteritis and colitis; ICD-10-CM index routes drug-induced colitis here

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Colitis	6,177	37,176	14	High
Immune-mediated enterocolitis	3,100	3,480	573.8	High
Enterocolitis	1,132	5,635	17.7	High
Autoimmune colitis	547	635	437.2	High
Colitis microscopic	207	4,632	3.3	Moderate

Look-alikes: Infectious colitis (C. difficile, CMV), pre-existing IBD flare, ischemic colitis, other drug diarrhea (e.g., concurrent TKI or chemotherapy).

Hepatitis (HEPATITIS)

Organ system: Gastrointestinal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Hepatitis (new in v6.0)	Symptomatic liver dysfunction; compensated cirrhosis;	High	New in CTCAE v6.0; graded hepatitis term

CTCAE v6.0 term	Grade 3	Confidence	Rationale
	hospitalization or prolongation of existing hospitalization indicated		

ICD-10-CM	Description	Billable codes	Confidence	Rationale
K75.4	Autoimmune hepatitis	1	Moderate	Autoimmune hepatitis; also used for ICI hepatitis but describes a distinct disease
K75.9	Inflammatory liver disease, unspecified	1	Low	Inflammatory liver disease, unspecified
K71.2	Toxic liver disease with acute hepatitis	1	High	Toxic liver disease with acute hepatitis
K71.1*	Toxic liver disease with hepatic necrosis	2	Moderate	Toxic liver disease with hepatic necrosis
K71.6	Toxic liver disease with hepatitis, not elsewhere classified	1	High	Toxic liver disease with hepatitis NEC
K71.9	Toxic liver disease, unspecified	1	Moderate	Toxic liver disease, unspecified

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Hepatitis	2,681	25,255	8.4	High
Immune-mediated hepatic disorder	1,830	1,887	2258	High
Immune-mediated hepatitis	1,814	1,938	1028.9	High
Drug-induced liver injury	1,357	30,439	3.3	High
Hepatic cytolysis	1,331	12,060	8.7	Moderate
Autoimmune hepatitis	1,187	6,833	14.8	High

Look-alikes: Viral hepatitis or HBV reactivation, hepatic metastases or progression, other hepatotoxic drugs (acetaminophen, azoles, TKIs), alcohol, biliary obstruction.

Pancreatitis / asymptomatic lipase elevation (PANCREATITIS)

Organ system: Gastrointestinal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Pancreatitis	Severe pain; vomiting; medical intervention indicated (e.g., analgesia, nutritional support)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
K85.9*	Acute pancreatitis, unspecified	3	Moderate	Acute pancreatitis, unspecified
K85.3*	Drug induced acute pancreatitis	3	High	Drug-induced acute pancreatitis
K85.8*	Other acute pancreatitis	3	Moderate	Other acute pancreatitis (includes autoimmune acute pancreatitis)

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Pancreatitis	1,293	52,220	1.8	High
Pancreatitis acute	410	22,020	1.3	High
Immune-mediated pancreatitis	355	364	2774.2	High
Autoimmune pancreatitis	151	498	30.6	High
Oedematous pancreatitis	30	551	4	Moderate

Look-alikes: Gallstone or alcohol pancreatitis, hypertriglyceridemia, pancreatic metastases; isolated enzyme rise without symptoms is common and often not treated.

Pneumonitis (PNEUMONITIS)

Organ system: Pulmonary

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Pneumonitis	Severe symptoms; oxygen indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term; most common fatal ICI toxicity in some series

ICD-10-CM	Description	Billable codes	Confidence	Rationale
J70.2	Acute drug-induced interstitial lung disorders	1	High	Acute drug-induced interstitial lung disorders
J70.3	Chronic drug-induced interstitial lung disorders	1	High	Chronic drug-induced interstitial lung disorders
J84.116	Cryptogenic organizing pneumonia	1	Moderate	Cryptogenic organizing pneumonia; organizing pneumonia is an ICI pattern
J70.4	Drug-induced interstitial lung disorders, unspecified	1	High	Drug-induced interstitial lung disorders, unspecified
J84.9	Interstitial pulmonary disease, unspecified	1	Moderate	Interstitial pulmonary disease, unspecified
J84.89	Other specified interstitial pulmonary diseases	1	Moderate	Other specified ILD

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Pneumonitis	6,132	26,681	21	High
Interstitial lung disease	5,821	47,680	9.8	High
Immune-mediated lung disease	1,618	1,725	1063.5	High
Organising pneumonia	439	4,843	7	High
Immune-mediated pneumonitis	220	234	1105.2	High
Autoimmune lung disease	80	114	165.5	High

Look-alikes: Infectious pneumonia (including PJP), radiation pneumonitis, lymphangitic tumor spread, pulmonary edema, pulmonary embolism, COPD exacerbation.

Thyroiditis / primary hypothyroidism (HYPOTHYROIDISM)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Hypothyroidism	Severe symptoms; hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term
Thyroiditis (new in v6.0)	Hospitalization indicated	High	New in v6.0; captures the destructive thyroiditis that precedes hypothyroidism

ICD-10-CM	Description	Billable codes	Confidence	Rationale
E06.3	Autoimmune thyroiditis	1	Moderate	Autoimmune thyroiditis; also pre-existing Hashimoto
E06.4	Drug-induced thyroiditis	1	High	Drug-induced thyroiditis
E03.2	Hypothyroidism due to medicaments and other exogenous substances	1	High	Hypothyroidism due to medicaments and other exogenous substances
E03.9	Hypothyroidism, unspecified	1	Moderate	Hypothyroidism, unspecified; very common at baseline
E06.5	Other chronic thyroiditis	1	Low	Other chronic thyroiditis
E03.8	Other specified hypothyroidism	1	Moderate	Other specified hypothyroidism
E06.9	Thyroiditis, unspecified	1	Moderate	Thyroiditis, unspecified

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Thyroiditis	1,281	2,941	54.3	High
Immune-mediated hypothyroidism	1,046	1,061	4904.5	High
Immune-mediated thyroiditis	458	463	6442.4	High
Autoimmune thyroiditis	394	3,715	8.3	High
Hypothyroidism	199	1,971	7.9	High
Autoimmune hypothyroidism	138	231	104.4	High

Look-alikes: Pre-existing hypothyroidism, post-surgical or post-radiation hypothyroidism, sick euthyroid syndrome, central hypothyroidism from hypophysitis (low TSH).

Thyroiditis / hyperthyroidism (HYPERTHYROIDISM)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Hyperthyroidism	Severe symptoms; hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term; usually transient thyrotoxic phase of thyroiditis
Thyroiditis (new in	Hospitalization indicated	High	New in v6.0

CTCAE v6.0 term	Grade 3	Confidence	Rationale
v6.0)			

ICD-10-CM	Description	Billable codes	Confidence	Rationale
E06.2	Chronic thyroiditis with transient thyrotoxicosis	1	Moderate	Chronic thyroiditis with transient thyrotoxicosis
E06.4	Drug-induced thyroiditis	1	High	Drug-induced thyroiditis
E05.8*	Other thyrotoxicosis	2	Moderate	Other thyrotoxicosis
E06.1	Subacute thyroiditis	1	Low	Subacute thyroiditis; painful viral thyroiditis is a look-alike
E05.9*	Thyrotoxicosis, unspecified	2	Moderate	Thyrotoxicosis, unspecified

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Hyperthyroidism	2,545	14,275	15.3	High
Thyroiditis	1,281	2,941	54.3	High
Immune-mediated thyroiditis	458	463	6442.4	High
Autoimmune thyroiditis	394	3,715	8.3	High
Immune-mediated hyperthyroidism	231	236	3249.3	High
Silent thyroiditis	100	142	167.5	High

Look-alikes: Iodinated contrast-induced thyrotoxicosis, toxic nodules, exogenous levothyroxine excess, viral subacute thyroiditis.

Hypophysitis / hypopituitarism (HYPOPHYSITIS)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Hypophysitis	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term; typical of anti-CTLA-4
Hypopituitarism	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
E23.7	Disorder of pituitary gland, unspecified	1	Low	Disorder of pituitary gland, unspecified

ICD-10-CM	Description	Billable codes	Confidence	Rationale
E23.1	Drug-induced hypopituitarism	1	High	Drug-induced hypopituitarism
E23.0	Hypopituitarism	1	Moderate	Hypopituitarism (other)
E23.6	Other disorders of pituitary gland	1	High	Other disorders of pituitary gland; ICD-10-CM index routes hypophysitis here

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Hypophysitis	1,855	1,986	995.9	High
Hypopituitarism	929	1,720	82.6	High
Adrenocorticotropic hormone deficiency	760	861	529.2	High
Hypothalamo-pituitary disorder	441	1,425	31.5	Moderate
Immune-mediated hypophysitis	416	423	4179.7	High
Lymphocytic hypophysitis	219	273	285.2	High

Look-alikes: Pituitary metastasis or adenoma, exogenous glucocorticoid suppression of the HPA axis, primary adrenal insufficiency.

Primary adrenal insufficiency (adrenalitis) (ADRENAL_INSUFFICIENCY)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Adrenal insufficiency	Severe symptoms; hospitalization indicated	High	Graded term; covers primary and secondary, so category needs context

ICD-10-CM	Description	Billable codes	Confidence	Rationale
E27.3	Drug-induced adrenocortical insufficiency	1	Moderate	Drug-induced adrenocortical insufficiency; also used for steroid withdrawal
E27.49	Other adrenocortical insufficiency	1	Low	Other adrenocortical insufficiency
E27.1	Primary adrenocortical insufficiency	1	Moderate	Primary adrenocortical insufficiency
E27.40	Unspecified adrenocortical insufficiency	1	Low	Unspecified adrenocortical insufficiency

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Adrenal insufficiency	3,896	11,815	34.6	High
Immune-mediated adrenal insufficiency	739	745	8662.5	High
Addison^s disease	144	987	12	Moderate
Primary adrenal insufficiency	94	294	33.1	High

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Adrenalitis	75	105	175.8	High

Look-alikes: Secondary adrenal insufficiency from hypophysitis (see HYPOPHYSITIS), glucocorticoid withdrawal, adrenal metastases or hemorrhage.

Immune checkpoint inhibitor diabetes (insulin-deficient) (DIABETES)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Diabetes mellitus (new in v6.0)	Insulin therapy initiated or increased; hospitalization indicated	High	New in v6.0

ICD-10-CM	Description	Billable codes	Confidence	Rationale
E09*	Drug or chemical induced diabetes mellitus	86	Moderate	Drug-induced diabetes; but most often steroid-induced hyperglycemia
E13*	Other specified diabetes mellitus	86	Moderate	Other specified diabetes
E10*	Type 1 diabetes mellitus	85	High	Type 1 diabetes; ICI diabetes is insulin-deficient and usually coded here
E10.1*	Type 1 diabetes mellitus with ketoacidosis	2	High	Type 1 diabetes with ketoacidosis; common presentation

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Type 1 diabetes mellitus	1,601	5,929	26	High
Diabetes mellitus	1,555	75,562	1.5	High
Diabetic ketoacidosis	1,461	23,655	4.6	High
Fulminant type 1 diabetes mellitus	838	1,007	348.7	High
Diabetes mellitus inadequate control	166	17,478	0.7	High
Diabetes insipidus	96	2,391	2.9	High

Look-alikes: Steroid-induced hyperglycemia (very common during irAE treatment), pre-existing type 2 diabetes, pancreatogenic diabetes.

Myocarditis (MYOCARDITIS)

Organ system: Cardiovascular

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Myocarditis	Severe with symptoms at rest or with minimal activity or exertion; intervention indicated; new onset of symptoms	High	Graded term; high fatality irAE

ICD-10-CM	Description	Billable codes	Confidence	Rationale
I40.9	Acute myocarditis, unspecified	1	High	Acute myocarditis, unspecified
I40.1	Isolated myocarditis	1	High	Isolated myocarditis
I51.4	Myocarditis, unspecified	1	Moderate	Myocarditis, unspecified
I40.8	Other acute myocarditis	1	High	Other acute myocarditis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Myocarditis	3,053	11,175	26.4	High
Immune-mediated myocarditis	1,441	1,519	1299.3	High
Myocarditis-myositis-myasthenia gravis overlap syndrome	174	179	2447.5	High
Autoimmune myocarditis	149	216	156.4	High

Look-alikes: Acute coronary syndrome, viral myocarditis, takotsubo, cardiotoxicity from anthracyclines or HER2 therapy; often overlaps with myositis and myasthenia (see MYOSITIS, MYASTHENIA).

Pericarditis / pericardial effusion (PERICARDITIS)

Organ system: Cardiovascular

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Pericarditis	Pericarditis with physiologic consequences (e.g., pericardial constriction)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
I30.0	Acute nonspecific idiopathic pericarditis	1	Moderate	Acute nonspecific idiopathic pericarditis
I30.9	Acute pericarditis, unspecified	1	High	Acute pericarditis, unspecified
I31.9	Disease of pericardium, unspecified	1	Low	Disease of pericardium, unspecified
I30.8	Other forms of acute pericarditis	1	High	Other forms of acute pericarditis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Pericarditis	407	16,450	1.8	High
Immune-mediated pericarditis	52	54	1828.6	High
Autoimmune pericarditis	12	20	105.5	High

Look-alikes: Malignant pericardial effusion (very common in lung cancer), radiation pericarditis, infection, uremia.

Vasculitis (VASCULITIS)

Organ system: Cardiovascular

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Vasculitis	Severe symptoms, medical intervention indicated (e.g., steroids)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
I77.6	Arteritis, unspecified	1	Moderate	Arteritis, unspecified
M31.5	Giant cell arteritis with polymyalgia rheumatica	1	Moderate	Giant cell arteritis with polymyalgia rheumatica
M31.9	Necrotizing vasculopathy, unspecified	1	Moderate	Necrotizing vasculopathy, unspecified
M31.6	Other giant cell arteritis	1	Moderate	Other giant cell arteritis
M31.8	Other specified necrotizing vasculopathies	1	Moderate	Other specified necrotizing vasculopathies
L95*	Vasculitis limited to skin, not elsewhere classified	4	Moderate	Vasculitis limited to skin

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Vasculitis	260	11,511	1.6	Moderate
Cutaneous vasculitis	88	3,524	1.8	Moderate
Giant cell arteritis	49	797	4.6	Moderate
Central nervous system vasculitis	40	482	6.4	Moderate
Temporal arteritis	34	927	2.7	Moderate
Hypersensitivity vasculitis	29	2,622	0.8	Moderate

Look-alikes: Infection, drug hypersensitivity vasculitis from other drugs, paraneoplastic vasculitis.

Nephritis (acute interstitial / glomerulonephritis) (NEPHRITIS)

Organ system: Renal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Glomerulonephritis (new in v6.0)	IV therapy indicated; immunosuppressant indicated; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	New in v6.0; less common with ICIs
Tubulointerstitial nephritis (new in v6.0)	IV therapy indicated; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity	High	New in v6.0; the typical ICI renal lesion

CTCAE v6.0 term	Grade 3	Confidence	Rationale
	(pediatric)		

ICD-10-CM	Description	Billable codes	Confidence	Rationale
N00*	Acute nephritic syndrome	13	Low	Acute nephritic syndrome
N14.1*	Nephropathy induced by other drugs, medicaments and biological substances	2	High	Nephropathy induced by other drugs, medicaments and biological substances
N12	Tubulo-interstitial nephritis, not specified as acute or chronic	1	Moderate	Tubulo-interstitial nephritis, not specified as acute or chronic
N05*	Unspecified nephritic syndrome	11	Low	Unspecified nephritic syndrome

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Tubulointerstitial nephritis	1,834	18,485	7.7	High
Nephritis	869	3,266	25.5	High
Nephropathy toxic	430	10,992	2.9	Moderate
Immune-mediated nephritis	412	447	827.9	High
Nephrotic syndrome	403	6,761	4.5	Moderate
Immune-mediated renal disorder	372	386	1868.8	High

Look-alikes: Prerenal AKI (dehydration from colitis), contrast nephropathy, other nephrotoxins (PPIs, NSAIDs, platinum), obstruction, pyelonephritis.

Inflammatory dermatitis / rash / pruritus (DERM_INFLAMMATORY)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Dermatitis psoriasiform (new in v6.0)	IV intervention indicated	High	New in v6.0; characteristic ICI pattern
Eczematous dermatitis (new in v6.0)	IV intervention indicated	High	New in v6.0
Erythroderma	Erythema covering >90% BSA with associated symptoms (e.g., pruritus or tenderness); limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	Severe diffuse dermatitis
Exfoliative dermatitis (new in v6.0)	IV intervention indicated	Moderate	New in v6.0
Lichenoid dermatitis (new in v6.0)	IV intervention indicated	High	New in v6.0; characteristic ICI pattern
Papulovesicular rash (new in v6.0)	Covering >50% BSA; moderate or severe symptoms	Moderate	New in v6.0

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Rash maculo-papular	Macules/papules covering >50% BSA; moderate or severe symptoms	High	Most common ICI skin irAE

ICD-10-CM	Description	Billable codes	Confidence	Rationale
L30.9	Dermatitis, unspecified	1	Low	Dermatitis, unspecified
L26	Exfoliative dermatitis	1	Moderate	Exfoliative dermatitis
L27.0	Generalized skin eruption due to drugs and medicaments taken internally	1	High	Generalized skin eruption due to drugs taken internally
L43.2	Lichenoid drug reaction	1	High	Lichenoid drug reaction
L27.1	Localized skin eruption due to drugs and medicaments taken internally	1	High	Localized skin eruption due to drugs taken internally

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Rash	9,000	427,888	1.5	High
Rash maculo-papular	1,402	21,868	4.8	High
Drug eruption	1,110	17,396	4.8	Moderate
Rash pruritic	1,014	53,237	1.4	High
Immune-mediated dermatitis	891	924	1899	High
Dermatitis	766	20,447	2.7	High

Look-alikes: EGFR inhibitor acneiform rash, radiation dermatitis, other drug eruptions (antibiotics), infection (scabies, fungal), xerosis.

Bullous dermatoses (bullous pemphigoid) (DERM_BULLOUS)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Dermatosis bullous (new in v6.0)	Blisters covering >50% BSA; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	New in v6.0; bullous pemphigoid is the typical ICI entity

ICD-10-CM	Description	Billable codes	Confidence	Rationale
L13.9	Bullous disorder, unspecified	1	Low	Bullous disorder, unspecified
L12.0	Bullous pemphigoid	1	High	Bullous pemphigoid
L12.8	Other pemphigoid	1	Moderate	Other pemphigoid
L12.9	Pemphigoid, unspecified	1	Moderate	Pemphigoid, unspecified

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Pemphigoid	1,279	7,314	14.9	High
Dermatitis bullous	419	7,100	4.4	High
Mucous membrane pemphigoid	19	103	15.9	High

Look-alikes: Bullous drug eruptions from other drugs, SJS/TEN (see DERM_SCAR), pemphigus, friction or edema blisters.

Severe cutaneous adverse reactions (SJS/TEN, DRESS, AGEP) (DERM_SCAR)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Erythema multiforme	Associated with oral or genital erosions	Moderate	Graded term
Pustular drug eruption (new in v6.0)	Severe symptoms; IV antibiotic or steroid indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	New in v6.0; AGEP
Stevens-Johnson syndrome	Skin sloughing covering <10% BSA	High	Graded term
Toxic epidermal necrolysis	-	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D72.12	Drug rash with eosinophilia and systemic symptoms syndrome	1	High	Drug rash with eosinophilia and systemic symptoms (DRESS)
L51.9	Erythema multiforme, unspecified	1	Moderate	Erythema multiforme, unspecified
L51.0	Nonbullous erythema multiforme	1	Moderate	Nonbullous erythema multiforme
L51.8	Other erythema multiforme	1	Moderate	Other erythema multiforme
L51.1	Stevens-Johnson syndrome	1	High	Stevens-Johnson syndrome
L51.3	Stevens-Johnson syndrome-toxic epidermal necrolysis overlap syndrome	1	High	SJS-TEN overlap
L51.2	Toxic epidermal necrolysis [Lyell]	1	High	Toxic epidermal necrolysis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Stevens-johnson syndrome	1,003	21,990	3.4	High
Toxic epidermal necrolysis	974	14,884	4.9	High
Erythema multiforme	687	9,202	5.7	High
Drug reaction with eosinophilia and systemic symptoms	324	25,217	0.9	High
Acute generalised exanthematous pustulosis	93	7,636	0.9	High

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Palmoplantar pustulosis	9	504	1.3	High

Look-alikes: SCAR from co-medications (allopurinol, sulfonamides, anticonvulsants, antibiotics) is often more likely than from the ICI.

Vitiligo / depigmentation (VITILIGO)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Skin hypopigmentation	-	High	CTCAE term covering vitiligo

ICD-10-CM	Description	Billable codes	Confidence	Rationale
L81.5	Leukoderma, not elsewhere classified	1	Moderate	Leukoderma, NEC
L81.6	Other disorders of diminished melanin formation	1	Low	Other disorders of diminished melanin formation
L80	Vitiligo	1	High	Vitiligo

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Vitiligo	508	2,275	20.2	High
Leukoderma	119	197	107.3	High
Skin depigmentation	43	1,341	2.3	High
Skin hypopigmentation	40	1,731	1.7	High
Retinal depigmentation	6	302	1.4	High

Look-alikes: Pre-existing vitiligo, post-inflammatory hypopigmentation.

Inflammatory arthritis (ARTHRITIS)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Arthritis	Severe pain associated with signs of inflammation, erythema, or joint swelling; irreversible joint damage; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
M13.8*	Other specified arthritis	23	Moderate	Other specified arthritis
M13.80	Other specified arthritis, unspecified site	1	Moderate	Other specified arthritis, unspecified site

ICD-10-CM	Description	Billable codes	Confidence	Rationale
M13.0	Polyarthritis, unspecified	1	Moderate	Polyarthritis, unspecified
M06.00	Rheumatoid arthritis without rheumatoid factor, unspecified site	1	Moderate	Seronegative rheumatoid arthritis, unspecified site
M06.9	Rheumatoid arthritis, unspecified	1	Moderate	Rheumatoid arthritis, unspecified; used for ICI inflammatory arthritis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Arthritis	1,215	77,756	1.1	High
Rheumatoid arthritis	890	111,584	0.6	High
Polyarthritis	430	5,184	6.4	High
Immune-mediated arthritis	350	365	1641.1	High
Autoimmune arthritis	82	153	81.2	High
Seronegative arthritis	65	1,108	4.4	High

Look-alikes: Osteoarthritis, gout or pseudogout, septic arthritis, pre-existing rheumatoid arthritis flare, bone metastases.

Polymyalgia rheumatica-like syndrome (PMR)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Musculoskeletal and connective tissue disorder - Other, specify	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE has no PMR term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
M31.5	Giant cell arteritis with polymyalgia rheumatica	1	Moderate	Giant cell arteritis with PMR
M35.3	Polymyalgia rheumatica	1	High	Polymyalgia rheumatica

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Polymyalgia rheumatica	363	3,074	9.4	High

Look-alikes: Myositis (CK elevated), hypothyroid myopathy, bone metastases, statin myalgia.

Sicca syndrome / Sjogren-like (SICCA)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Salivary duct inflammation	Acute salivary gland necrosis; severe secretion-induced symptoms (e.g., thick saliva/oral secretions or gagging); tube feeding or TPN indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	Sialadenitis

ICD-10-CM	Description	Billable codes	Confidence	Rationale
K11.2*	Sialoadenitis	4	Moderate	Sialoadenitis
M35.0*	Sjogren syndrome	13	High	Sicca syndrome (Sjogren)

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Sjogren^s syndrome	336	4,414	5.8	High
Sialoadenitis	126	1,340	7.3	Moderate
Sicca syndrome	34	172	17.3	High
Noninfective sialoadenitis	4	157	1.8	Moderate

Look-alikes: Radiation xerostomia (head and neck RT), anticholinergic drugs, pre-existing Sjogren.

Myositis (MYOSITIS)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Myositis	Pain associated with severe weakness; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term; overlaps with myocarditis and myasthenia

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G72.4*	Inflammatory and immune myopathies, not elsewhere classified	2	Moderate	Inflammatory and immune myopathies, NEC
M60.9	Myositis, unspecified	1	High	Myositis, unspecified
M33.1*	Other dermatomyositis	5	Moderate	Other dermatomyositis
M60.8*	Other myositis	24	High	Other myositis
M33.2*	Polymyositis	4	Moderate	Polymyositis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Myositis	1,859	8,854	18.7	High
Immune-mediated myositis	732	2,603	27.5	High

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Myopathy	292	8,551	2.5	Moderate
Myocarditis-myositis-myasthenia gravis overlap syndrome	174	179	2447.5	High
Polymyositis	126	1,744	5.5	High
Autoimmune myositis	86	452	16.5	High

Look-alikes: Statin myopathy, steroid myopathy, hypothyroid myopathy, paraneoplastic myositis; screen for concurrent myocarditis and myasthenia.

Myasthenia gravis / myasthenic syndrome (MYASTHENIA)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Myasthenia gravis	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G70.80	Lambert-Eaton syndrome, unspecified	1	Low	Lambert-Eaton syndrome, unspecified
G70.01	Myasthenia gravis with (acute) exacerbation	1	High	Myasthenia gravis with exacerbation
G70.00	Myasthenia gravis without (acute) exacerbation	1	High	Myasthenia gravis without exacerbation
G70.1	Toxic myoneural disorders	1	Moderate	Toxic myoneural disorders

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Myasthenia gravis	1,294	7,772	14	High
Immune-mediated myasthenia gravis	289	301	1693.8	High
Myasthenic syndrome	260	752	37.2	High
Myocarditis-myositis-myasthenia gravis overlap syndrome	174	179	2447.5	High
Myasthenia gravis crisis	78	1,415	4.1	High
Ocular myasthenia	48	340	11.6	High

Look-alikes: Pre-existing MG, Lambert-Eaton (paraneoplastic, small cell lung cancer), myositis with ocular involvement, brainstem lesions.

Guillain-Barre syndrome / inflammatory polyradiculoneuropathy (GBS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Guillain-Barre syndrome	Severe symptoms; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G61.81	Chronic inflammatory demyelinating polyneuritis	1	Moderate	Chronic inflammatory demyelinating polyneuritis
G61.0	Guillain-Barre syndrome	1	High	Guillain-Barre syndrome
G61.9	Inflammatory polyneuropathy, unspecified	1	Low	Inflammatory polyneuropathy, unspecified
G61.89	Other inflammatory polyneuropathies	1	Moderate	Other inflammatory polyneuropathies

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Guillain-barre syndrome	517	4,761	8.6	High
Demyelinating polyneuropathy	82	1,016	6.2	High
Chronic inflammatory demyelinating polyradiculoneuropathy	75	1,400	4	High
Miller fisher syndrome	43	223	16.8	High
Polyradiculoneuropathy	18	39	60.3	High
Axonal and demyelinating polyneuropathy	16	92	14.8	High

Look-alikes: Chemotherapy neuropathy, leptomeningeal disease, spinal cord compression, critical illness polyneuropathy.

Peripheral, cranial, and autonomic neuropathy (PERIPHERAL_NEUROPATHY)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Autonomic neuropathy (new in v6.0)	Severe symptoms; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	New in v6.0
Facial nerve disorder	Severe symptoms; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	Facial palsy is a reported ICI neuropathy

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G51.0	Bell's palsy	1	Moderate	Bell palsy
G90.9	Disorder of the autonomic nervous system, unspecified	1	Low	Disorder of autonomic nervous system, unspecified

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G62.0	Drug-induced polyneuropathy	1	Moderate	Drug-induced polyneuropathy; usually chemotherapy (platinum, taxane)
G62.9	Polyneuropathy, unspecified	1	Low	Polyneuropathy, unspecified

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Facial paralysis	374	8,372	3.3	Moderate
Immune-mediated neuropathy	119	144	334.8	High

Look-alikes: Chemotherapy-induced neuropathy (very common in combination regimens), diabetic neuropathy, B12 deficiency, tumor compression.

Aseptic meningitis (ASEPTIC_MENINGITIS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Aseptic meningitis (new in v6.0)	Steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	New in v6.0

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G03.8	Meningitis due to other specified causes	1	Moderate	Meningitis due to other specified causes
G03.9	Meningitis, unspecified	1	Low	Meningitis, unspecified
G03.0	Nonpyogenic meningitis	1	High	Nonpyogenic meningitis (aseptic)

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Meningitis	387	6,596	4.4	Moderate
Meningitis aseptic	377	4,416	6.6	High
Meningitis noninfective	9	62	11.9	High

Look-alikes: Infectious meningitis, leptomeningeal carcinomatosis, drug-induced aseptic meningitis (NSAIDs, IVIG).

Encephalitis / meningoencephalitis (ENCEPHALITIS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Encephalitis (new in v6.0)	IV antibiotic, antifungal, or antiviral intervention indicated; steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	New in v6.0

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Encephalomyelitis (new in v6.0)	IV antibiotic, antifungal, or antiviral intervention indicated; steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	New in v6.0
Meningoencephalitis (new in v6.0)	IV antibiotic, antifungal, or antiviral intervention indicated; steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	New in v6.0

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G04.00	Acute disseminated encephalitis and encephalomyelitis, unspecified	1	Moderate	Acute disseminated encephalitis, unspecified
G04.90	Encephalitis and encephalomyelitis, unspecified	1	Moderate	Encephalitis and encephalomyelitis, unspecified
G04.81	Other encephalitis and encephalomyelitis	1	High	Other encephalitis and encephalomyelitis (includes autoimmune encephalitis)
G04.89	Other myelitis	1	Moderate	Other myelitis/encephalitis
G04.02	Postimmunization acute disseminated encephalitis, myelitis and encephalomyelitis	1	Moderate	Postimmunization ADEM; check index

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Encephalitis	1,094	6,807	13.5	High
Immune-mediated encephalitis	457	489	1004.4	High
Encephalitis autoimmune	353	1,276	26.9	High
Limbic encephalitis	81	276	29.2	High
Noninfective encephalitis	40	706	4.2	High
Meningoencephalitis herpetic	23	448	3.8	High

Look-alikes: HSV or other viral encephalitis, brain metastases, PRES, metabolic encephalopathy, ICANS after CAR-T.

Transverse myelitis (MYELITIS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Myelitis	Severe weakness or sensory loss; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	Graded term (Infections SOC in CTCAE, but non-infectious myelitis also graded here)

ICD-10-CM	Description	Billable codes	Confidence	Rationale
G37.3	Acute transverse myelitis in demyelinating disease of central nervous system	1	High	Acute transverse myelitis in demyelinating disease
G04.91	Myelitis, unspecified	1	Moderate	Myelitis, unspecified

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Myelitis	100	1,533	4.9	High
Myelitis transverse	98	1,391	5.3	High
Immune-mediated myelitis	15	23	131.9	High

Look-alikes: Epidural metastasis or cord compression, radiation myelopathy, infection, multiple sclerosis or NMOSD.

Uveitis / scleritis / episcleritis (OCULAR_INFLAMMATION)

Organ system: Ocular

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Scleral disorder	Symptomatic with marked decrease in visual acuity (best corrected visual acuity worse than 20/40 or more than 3 lines of decreased vision from known baseline, up to 20/200); limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	Scleritis/episcleritis
Uveitis	Anterior uveitis with 3+ or greater cells; intermediate posterior or pan-uveitis	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
H30*	Chorioretinal inflammation	56	Moderate	Chorioretinal inflammation (posterior uveitis)
H15.1*	Episcleritis	12	Moderate	Episcleritis
H20*	Iridocyclitis	38	High	Iridocyclitis (anterior uveitis)
H44.11*	Panuveitis	4	High	Panuveitis
H15.0*	Scleritis	28	Moderate	Scleritis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Uveitis	608	13,330	3.4	High
Immune-mediated uveitis	174	178	3059.4	High
Iridocyclitis	143	3,086	3.4	High
Vogt-koyanagi-harada disease	90	212	51.9	High
Iritis	56	3,000	1.3	High

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Chorioretinitis	38	635	4.5	High

Look-alikes: Infectious uveitis (HSV, CMV, syphilis), endophthalmitis, conjunctivitis, MEK inhibitor retinopathy (combination BRAF/MEK).

Optic neuritis (OPTIC_NEURITIS)

Organ system: Ocular

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Optic nerve disorder	Marked decrease in visual acuity (best corrected visual acuity worse than 20/40 or more than 3 lines of decreased vision from known baseline, up to 20/200)	Moderate	Broader term
Optic neuritis (new in v6.0)	Severe or medically significant but not immediately sight-threatening; decrease in visual acuity (best corrected visual acuity worse than 20/40 or more than 3 lines of decreased vision from known baseline, up to 20/200) in the affected eye; oral steroids indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	New in v6.0

ICD-10-CM	Description	Billable codes	Confidence	Rationale
H47.01*	Ischemic optic neuropathy	4	Moderate	Ischemic optic neuropathy
H46*	Optic neuritis	12	High	Optic neuritis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Optic neuritis	264	9,000	2.1	High
Optic neuropathy	59	2,071	2.1	High
Immune-mediated optic neuritis	25	26	1758.3	High
Toxic optic neuropathy	3	425	0.5	High

Look-alikes: Ischemic optic neuropathy, papilledema, MS or NMOSD, compressive lesion.

Autoimmune hemolytic anemia (AIHA)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Hemolytic anemia (new in v6.0)	>2 units of PRBCs per day indicated	High	New in v6.0

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D59.10	Autoimmune hemolytic anemia, unspecified	1	Moderate	Autoimmune hemolytic anemia, unspecified
D59.12	Cold autoimmune hemolytic anemia	1	Moderate	Cold autoimmune hemolytic anemia
D59.0	Drug-induced autoimmune hemolytic anemia	1	High	Drug-induced autoimmune hemolytic anemia
D59.13	Mixed type autoimmune hemolytic anemia	1	Moderate	Mixed type AIHA
D59.19	Other autoimmune hemolytic anemia	1	Low	Other autoimmune hemolytic anemia
D59.11	Warm autoimmune hemolytic anemia	1	Moderate	Warm autoimmune hemolytic anemia

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Autoimmune haemolytic anaemia	422	3,366	10.1	High
Haemolytic anaemia	298	8,876	2.4	Moderate
Haemolysis	99	7,637	0.9	Moderate
Cold type haemolytic anaemia	18	229	6	High
Warm type haemolytic anaemia	10	94	8.4	High

Look-alikes: Microangiopathic hemolysis (TMA), G6PD deficiency with oxidant drugs, transfusion reaction, marrow infiltration.

Immune thrombocytopenia (ITP)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Thrombocytopenia (new in v6.0)	<50,000 - 10,000/mm3; <50.0 - 10.0 x 10^9/L; transfusion indicated	Moderate	New in v6.0 as a disorder term; also chemotherapy-related

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D69.3	Immune thrombocytopenic purpura	1	High	Immune thrombocytopenic purpura
D69.59	Other secondary thrombocytopenia	1	Low	Other secondary thrombocytopenia

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Immune thrombocytopenia	570	4,416	10.4	High

Look-alikes: Chemotherapy myelosuppression, marrow infiltration, HIT, DIC, TTP.

Aplastic anemia / marrow failure (APLASTIC_ANEMIA)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Aplastic anemia (new in v6.0)	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated	High	New in v6.0
Bone marrow hypocellular	Severely hypocellular or >50 - <=75% reduction cellularity from normal for age	Moderate	Marrow failure

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D60*	Acquired pure red cell aplasia [erythroblastopenia]	4	Moderate	Acquired pure red cell aplasia
D61.9	Aplastic anemia, unspecified	1	Moderate	Aplastic anemia, unspecified
D61.1	Drug-induced aplastic anemia	1	High	Drug-induced aplastic anemia
D61.811	Other drug-induced pancytopenia	1	Low	Other drug-induced pancytopenia
D61.89	Other specified aplastic anemias and other bone marrow failure syndromes	1	Moderate	Other specified aplastic anemias

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Bone marrow failure	400	20,194	1.4	High
Febrile bone marrow aplasia	328	4,133	6.1	High
Aplasia pure red cell	188	3,394	4.1	High
Aplastic anaemia	138	5,175	1.9	High
Autoimmune aplastic anaemia	5	13	44	High

Look-alikes: Chemotherapy myelosuppression, marrow infiltration by tumor, HLH.

Thrombotic microangiopathy (TTP / HUS) (TMA)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Hemolytic uremic syndrome	Laboratory findings with clinical consequences (e.g., renal insufficiency, petechiae)	High	Graded term
Thrombotic microangiopathy (new in v6.0)	Transfusion or IV intervention indicated	High	New in v6.0
Thrombotic thrombocytopenic purpura	Laboratory findings with clinical consequences (e.g., renal insufficiency, petechiae)	High	Graded term

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D59.3*	Hemolytic-uremic syndrome	4	Moderate	Hemolytic-uremic syndrome
M31.19	Other thrombotic microangiopathy	1	High	Other thrombotic microangiopathy (includes acquired TTP)
M31.10	Thrombotic microangiopathy, unspecified	1	High	Thrombotic microangiopathy, unspecified

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Thrombotic microangiopathy	260	9,560	2	High
Thrombotic thrombocytopenic purpura	97	3,485	2	High
Haemolytic uraemic syndrome	35	2,315	1.1	High
Renal-limited thrombotic microangiopathy	17	223	5.8	High
Atypical haemolytic uraemic syndrome	13	796	1.2	High
Pulmonary tumour thrombotic microangiopathy	10	75	10.8	High

Look-alikes: DIC, malignant hypertension, chemotherapy (gemcitabine, mitomycin) or calcineurin inhibitor TMA.

Hemophagocytic lymphohistiocytosis (HLH)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Secondary hemophagocytic lymphohistiocytosis (new in v6.0)	Severe or medically significant but not immediately life threatening; hospitalization or prolongation of existing hospitalization indicated	High	New in v6.0

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D76.1	Hemophagocytic lymphohistiocytosis	1	High	Hemophagocytic lymphohistiocytosis

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Haemophagocytic lymphohistiocytosis	849	7,131	9.5	High
Macrophage activation	2	34	4.4	High

Look-alikes: Infection- or lymphoma-associated HLH, sepsis, CRS.

Sarcoid-like granulomatous reaction (SARCOID_LIKE)

Organ system: Systemic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Immune system disorders - Other,	Severe or medically significant but not immediately life-threatening;	Moderate	No dedicated CTCAE term

CTCAE v6.0 term	Grade 3	Confidence	Rationale
specify	hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)		

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D86*	Sarcoidosis	13	Moderate	Sarcoidosis; ICI sarcoid-like reaction usually coded here

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Sarcoidosis	209	4,690	3.3	High
Pulmonary sarcoidosis	92	1,091	6.5	High
Cutaneous sarcoidosis	42	418	7.9	High
Sarcoidosis of lymph node	8	53	12.5	High
Cardiac sarcoidosis	7	228	2.2	High
Ocular sarcoidosis	4	119	2.4	High

Look-alikes: Disease progression (hilar nodes mimic metastasis), TB or fungal granulomas.

Immune-related adverse event, organ not specified (IRAE_UNSPECIFIED)

Organ system: Systemic

CTCAE v6.0 term	Grade 3	Confidence	Rationale
Autoimmune disorder	Autoimmune reactions involving major organ (e.g., colitis, anemia, myocarditis, kidney)	Moderate	Catch-all autoimmune term
Immune system disorders - Other, specify	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Low	Catch-all

ICD-10-CM	Description	Billable codes	Confidence	Rationale
D89.9	Disorder involving the immune mechanism, unspecified	1	Low	Disorder involving the immune mechanism, unspecified
D89.89	Other specified disorders involving the immune mechanism, not elsewhere classified	1	Low	Other specified disorders involving the immune mechanism

FAERS reaction term	ICI reports	All reports	PRR	Confidence
Immune-mediated adverse reaction	1,130	1,442	254.7	Moderate
Autoimmune disorder	757	9,445	6.1	Moderate
Immune-mediated endocrinopathy	136	140	2391.3	Moderate

Look-alikes: Use only when the organ is unknown; organ-specific categories take precedence.

6. Drug-cause companion codes

Code	Description	Role	Note
T45.1X5*	Adverse effect of antineoplastic and immunosuppressive drugs	companion	Adverse effect of antineoplastic and immunosuppressive drugs. The code used for ICI adverse effects before 2024-10-01; also covers chemotherapy and immunosuppressants.
T45.AX5*	Adverse effect of immune checkpoint inhibitors and immunostimulant drugs	companion	Adverse effect of immune checkpoint inhibitors and immunostimulant drugs. Added in FY2025 (effective 2024-10-01). Use for encounters on or after that date.
T45.AX1*	Poisoning by immune checkpoint inhibitors and immunostimulant drugs, accidental (unintentional)	lookalike	Poisoning (overdose), not an adverse effect at therapeutic dose; listed so analysts do not confuse it.

An ICD-10-CM condition code alone does not attribute the condition to a checkpoint inhibitor. Attribution in coded data depends on a drug-cause code on the same encounter: T45.AX5- for encounters on or after 1 October 2024, and T45.1X5- before that date. T45.1X5- also covers chemotherapy and immunosuppressants, so analyses before October 2024 cannot separate checkpoint inhibitor toxicity from chemotherapy toxicity by code alone.

7. Limitations

A code is not a diagnosis of an irAE. An ICD-10-CM code such as K52.1 (toxic gastroenteritis and colitis) says what the clinician recorded, not that an immune checkpoint inhibitor caused it. Attribution in records usually depends on a companion drug-cause code (T45.AX5- or T45.1X5-), clinical notes, and timing. irAE-Map supplies the code lists; it does not validate them against chart review. Sensitivity and positive predictive value of these code sets in real EHR data are unknown until someone measures them (see NEXT_STEPS).

The drug-cause code changed on 2024-10-01. T45.AX5- (adverse effect of immune checkpoint inhibitors and immunostimulant drugs) entered ICD-10-CM in FY2025. Before that date, ICI adverse effects were coded with T45.1X5-, which also covers chemotherapy and immunosuppressants. Analyses spanning that date must use both, and T45.1X5- alone cannot separate ICI from chemotherapy toxicity. T45.AX5- also covers "immunostimulant drugs" beyond checkpoint inhibitors.

Many codes are shared with non-immune causes. Codes rated Moderate or Low confidence (for example E03.9 hypothyroidism, unspecified; K52.9 noninfective colitis, unspecified) are common in patients who never received immunotherapy. Their usefulness depends on the analyst restricting to treated patients and a time window.

FAERS counts are reports, not patients or incidence. FAERS is a spontaneous reporting system. Counts reflect reporting behavior, duplicates, stimulated reporting after publications and

label changes, and missing data. They cannot estimate how often an irAE happens. ici_prr compares how often a term appears in reports naming an ICI versus all other reports; it is a descriptive over-representation measure, not a validated disproportionality signal (no confidence interval, no stratification, no deduplication).

openFDA cannot tie the reaction to the ICI within a report. The query finds reports in which any listed drug is one of the 13 ICIs (by harmonized generic name or by raw product name, generic or brand). It cannot require that the ICI is the *suspect* drug, because openFDA does not match drug-level fields within the same drug record. Some counted reports will list the ICI as a concomitant drug.

openFDA coverage of drug names is incomplete. On 2026-10-13, openFDA's harmonized generic-name field returned 0 reports for avelumab, tislelizumab and retifanlimab; we added raw product-name matching (generic and brand names) to recover them. Misspelled or foreign product names are still missed.

FAERS candidate terms come from keyword searches capped at 100 terms each (without an openFDA API key). Rare MedDRA terms outside the top 100 for every keyword can be missed. Re-running code/01_fetch.py with OPENFDA_API_KEY set raises the cap to 1,000. Report counts for the candidate terms themselves are exact (direct per-term queries where the keyword count was not exact).

Category boundaries are a clinical judgment. The categories follow the ASCO 2021 irAE guideline's organ groups, with additions. Other groupings (for example combining thyroid categories, or separating cholangitis from hepatitis) are equally defensible. Some terms legitimately map to more than one category (thyroiditis; the myocarditis-myositis-myasthenia overlap).

MedDRA licensing. CTCAE v6.0 terms and FAERS reaction terms are MedDRA terms. MedDRA is maintained by ICH and licensed through MSSO. irAE-Map republishes term names and the CTCAE LLT codes exactly as they appear in NCI's and FDA's free public releases, and does not redistribute the MedDRA hierarchy or MedDRA PT codes. [VERIFY] Users who need MedDRA codes or hierarchy should obtain a MedDRA license.

ICD-10-CM is US-specific and versioned. Codes are valid for FY2027 (2026-10-01 to 2027-09-30). Codes added, deleted or revised in later releases require a new irAE-Map version (v2.0 planned for FY2028). Other countries' ICD-10 modifications differ.

CTCAE grade text is copied verbatim. irAE-Map does not convert grades into code-based rules; ICD-10-CM codes do not carry CTCAE grade except G92.0- (ICANS), which is not an ICI irAE.

Candidates were proposed by rules and drafted with AI assistance. Search rules, code families and draft rationales were produced by Claude (Anthropic) from the public sources. Every mapping in the released table was reviewed and decided by the author; rows not decided are not released. Rules can still miss relevant codes or terms; the full candidate lists are published so others can check what was considered.

8. Use of AI assistance

Claude (Anthropic) wrote the data-processing code, the candidate search rules, and the draft role, confidence and rationale for candidates. The author made and checked every mapping decision, reviewed the code's outputs against the sources, and is responsible for the content.

9. Versioning and maintenance

irAE-Map uses semantic versioning. ICD-10-CM changes every 1 October; version 2.0, aligned to the FY2028 code files, is planned for October 2027 after CMS and CDC publish them. A Python package, iraemap, that applies the table to a user's codes is planned (version 0.1, March 2027). Corrections and suggestions are welcome through the repository's issue tracker.

10. How to cite

[VERIFY: name exactly as it should appear]. irAE-Map: codebook and mapping rationale, version 1.0.0. Zenodo. DOI: [pending first release].